

INTERNSHIP PROJECT REPORT

AegisAI Enterprise SIEM

An AI-Powered Security Information and
Event Management Platform

*Submitted in Partial Fulfillment of the
HCiC Cyber Security Internship Program Requirements*



STUDENT DETAILS	
Submitted By	Parag Patel
Intern ID	HCIC-1233
Email	paragpatel1020@gmail.com
Internship Domain	Digital Forensics & Cyber Security
Project Duration	HCIC Internship 2026

PROJECT INFORMATION	
Organization	HackHalt / HCiC Internship Program
Project Type	Major Internship Project
Technology	AI • SIEM • SOAR • SOC
Academic Year	2026

SUBMITTED TO
HackHalt Cyber Security Internship (HCiC)
2026



HACKHALT CYBER INTELLIGENCE COUNCIL

(हैकहॉल्ट साइबर इंटेलिजेंस काउंसिल)

CIN: U94990DL2025NPL459178 | PAN: AAICH3354B | MSME: UDYAM-DL-02-0107864

NITI AAYOG: DL/2025/0920083 | MSJE: DL/00051452



Date: 3 June, 2026

Ref. No.: HCIC-SI2026

INTERNSHIP LETTER

Dear Interns,

We are thrilled to offer you the position of **Cybersecurity Intern** at HackHalt Cyber Intelligence Council, starting **June 15, 2026**, for one month remotely.

Internship Details:

- **Role:** Cybersecurity Intern
- **Manager:** Piyush Sinha (Operation Associate)
- **Work Type:** Remote

Key Responsibilities:

1. Conduct cybersecurity research to identify potential vulnerabilities and enhance secure coding practices.
2. Analyze market trends in cybersecurity technologies and propose innovative solutions.
3. Assist in the implementation and monitoring of security measures to protect company data.
4. Develop and maintain technical documentation related to cybersecurity protocols and procedures.
5. Collaborate in the development of internal utility tools to improve system security and efficiency.

Terms & Recognition:

Complete the full month for an Internship Completion Certificate and maintain confidentiality regarding company data.

To accept, please sign and return by June 7, 2026. We look forward to having you on board!

Best Regards,

Piyush Sinha

Piyush Sinha



ACKNOWLEDGEMENT

I would like to express my sincere gratitude to HackHalt Intelligence Council (HCIC) for providing me with the opportunity to participate in the HCIC-SI2026 Cyber Security Internship. This internship provided an excellent platform to strengthen my practical knowledge in Security Information and Event Management (SIEM), digital forensics, threat detection, and incident response.

I am thankful to the mentors, reviewers, and the entire HCIC team for their continuous guidance, valuable feedback, and technical support throughout the project development process. Their encouragement motivated me to explore modern cybersecurity technologies and implement real-world security monitoring techniques.

I would also like to thank Gujarat University for encouraging practical learning and providing an environment that supports innovation and cybersecurity research.

Finally, I express my heartfelt gratitude to my family and friends for their continuous motivation, encouragement, and support during the successful completion of this project.

Submitted By

Parag Patel

HCIC-SI2026 Intern

ABSTRACT

Modern organizations generate thousands of security events every day, making it difficult for security analysts to identify real threats efficiently. Traditional Security Information and Event Management (SIEM) solutions often require extensive manual investigation, resulting in delayed incident response and increased security risks.

The **AegisAI Enterprise SIEM – Autonomous Agentic SIEM & Incident Response Platform** is designed to address these challenges by integrating real-time log collection, intelligent threat detection, behavioral analytics, MITRE ATT&CK mapping, threat intelligence, and automated incident response into a unified platform.

The system continuously monitors endpoint activities, security events, and network telemetry to identify suspicious behavior. It correlates events from multiple sources and provides analysts with real-time dashboards, alerts, incident timelines, and remediation capabilities. The platform follows a modular architecture, allowing scalability and easy integration with enterprise security tools.

This project demonstrates the practical implementation of modern SOC (Security Operations Center) workflows using technologies such as React, Node.js, Splunk Enterprise, Wazuh, Sysmon, Windows Event Logs, and MITRE ATT&CK. The solution improves threat visibility, reduces response time, and provides a centralized environment for security monitoring and incident management.

The project serves as a prototype of an intelligent enterprise SIEM platform capable of supporting cybersecurity professionals in detecting, investigating, and responding to security incidents more effectively.

Keywords:

SIEM, SOC, Cybersecurity, Threat Detection, Incident Response, Splunk Enterprise, Wazuh, Sysmon, MITRE ATT&CK, Threat Intelligence, Security Monitoring, Digital Forensics.

TABLE OF CONTENTS

Preliminary Pages

Cover Page	1
Internship Offer Letter	2
Acknowledgement	3
Abstract	4
Table of Contents	5
Table of Contents (Continued)	6
Table of Contents (Continued)	7

Chapter 1 – Executive Overview

1.1 Project Introduction	8
1.2 Project Objectives	8

Chapter 2 – Problem Statement

2.1 Background.....	9
2.2 Problem Identification.....	9
2.3 Need for Proposed Solution.....	9

Chapter 3 – Existing System

3.1 Existing Security Monitoring Approach	10
3.2 Limitations of Existing Systems.....	10
3.3 Summary	10

Chapter 4 – Proposed System

4.1 Proposed Solution	11
4.2 System Design Approach	11
4.3 Core Functional Components	11
4.4 Major Advantages	12

Chapter 5 – Project Objectives

5.1 Primary Objective	13
-----------------------------	----

5.2 Specific Objectives	13
-------------------------------	----

5.3 Expected Outcomes	13
-----------------------------	----

Chapter 6 – Technology Stack

6.1 Technology Overview	15
-------------------------------	----

6.2 Technology Stack.....	15
---------------------------	----

6.3 Core Technologies	16
-----------------------------	----

Chapter 7 – System Architecture

7.1 Architecture Overview.....	17
--------------------------------	----

7.2 Architecture Diagram	17
--------------------------------	----

7.3 Architecture Components	18
-----------------------------------	----

Chapter 8 – System Workflow

8.1 Workflow Overview	19
-----------------------------	----

8.2 Operational Workflow	19
--------------------------------	----

8.3 Workflow Stages	20
---------------------------	----

Chapter 9 – Project Modules

9.1 Module Overview	21
---------------------------	----

9.2 SOC Dashboard	22
-------------------------	----

9.3 Alert Center.....	24
-----------------------	----

9.4 AI Incident Analysis.....	26
-------------------------------	----

9.5 MITRE ATT&CK Mapping	28
--------------------------------	----

9.6 Threat Intelligence	30
-------------------------------	----

9.7 Global Threat Recon	32
-------------------------------	----

9.8 IOC Threat Lookup	34
-----------------------------	----

9.9 Case Management.....	36
--------------------------	----

9.10 Incident Timeline	38
------------------------------	----

9.11 Audit Reports	40
--------------------------	----

9.12 Security Analytics	42
-------------------------------	----

9.13 User Profile	44
-------------------------	----

9.14 SIEM Settings	46
9.15 About AegisAI	48
9.16 SOC Documentation.....	50
Chapter 25 – Testing & Validation	
25.1 Testing Overview	53
25.2 Testing Environment.....	53
25.3 Functional Testing Results	53
25.4 Performance Evaluation	54
25.5 Validation Summary.....	54
25.6 Summary	55
Chapter 26 – Results & Discussion	
26.1 Project Results	56
26.2 Project Acievements	56
26.3 Project Outcomes	56
26.4 Future scope	57
26.5 Conclusion.....	57
26.6 Skill Gained	58
Chapter 27 – References	
27.1 Books	59
27.2 Offical Documentation.....	59
27.3 Security Resources	59
27.4 Software Tools used	60
27. Citation Note.....	60

EXECUTIVE OVERVIEW

1.1 Project Introduction

The rapid increase in cyber threats has made continuous security monitoring an essential requirement for modern organizations. Security Operations Centers (SOCs) receive thousands of security events every day, making it difficult for analysts to identify genuine threats using traditional monitoring approaches. Manual log analysis often results in delayed incident response, alert fatigue, and increased security risks.

AegisAI Enterprise SIEM is an AI-assisted Security Information and Event Management (SIEM) platform designed to automate security monitoring, threat detection, event correlation, and incident response. The platform collects security telemetry from multiple sources, analyzes suspicious activities in real time, and provides security analysts with an intuitive dashboard for investigation and response.

1.2 Project Objective

The primary objective of this project is to develop an enterprise-grade SIEM platform capable of monitoring endpoint activities, detecting security threats, visualizing incidents, and assisting security analysts through automated workflows. The system aims to reduce response time, improve visibility across security events, and provide actionable intelligence for incident handling.

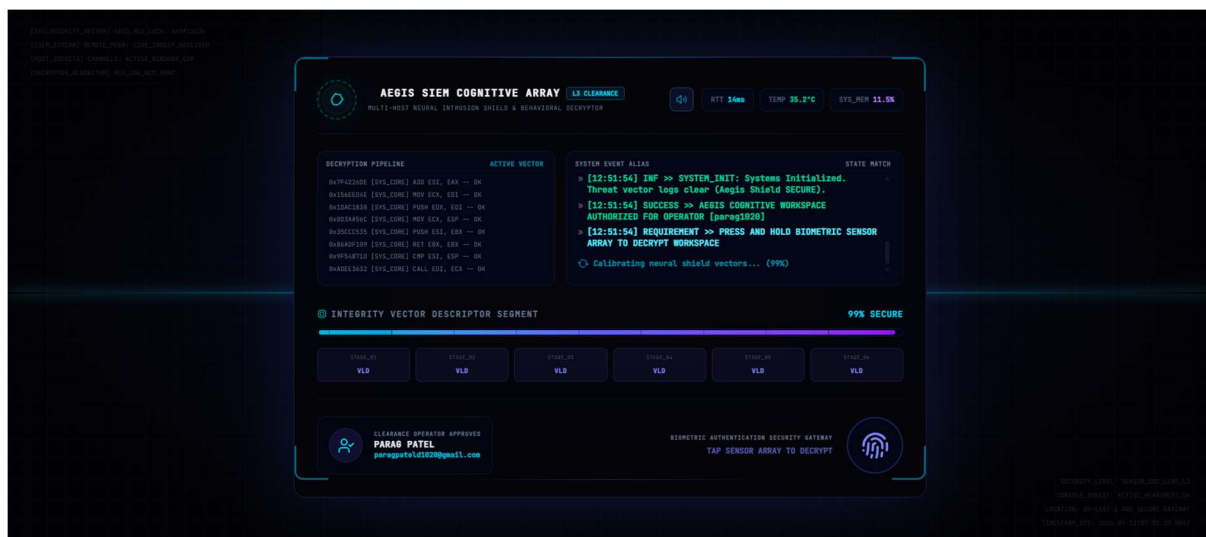


Figure 1 : AegisAI Enterprise SIEM Startup Interface

PROBLEM STATEMENT

2.1 Background

Modern organizations generate millions of security events every day from endpoints, servers, applications, firewalls, and cloud services. Security analysts often struggle to identify genuine threats because of the overwhelming volume of alerts, resulting in alert fatigue, delayed investigations, and increased cyber risk.

2.2 Problem Identification

Existing Problem	Impact
Manual Log Analysis	Slow Investigation
Multiple Security Tools	No Centralized Visibility
Alert Fatigue	Important Alerts Missed
Delayed Response	Higher Security Risk
Limited Automation	Increased Analyst Workload

2.3 Need for the Proposed Solution

To overcome these limitations, there is a need for an intelligent SIEM platform capable of collecting security events, correlating logs from multiple sources, detecting threats in real time, mapping attacks to the MITRE ATT&CK framework, and assisting analysts through automated incident response. AegisAI Enterprise SIEM is designed to address these challenges by providing centralized visibility, real-time analytics, and AI-assisted security operations.

EXISTING SYSTEM

3.1 Existing Security Monitoring Approach

Traditional Security Information and Event Management (SIEM) solutions primarily focus on collecting and storing security logs from multiple sources. Although these systems provide centralized log management, they often depend heavily on manual analysis by security analysts. As organizations continue to generate massive volumes of security events, identifying genuine threats becomes increasingly difficult and time-consuming.

Most traditional monitoring platforms lack intelligent event correlation, automated response mechanisms, contextual threat intelligence, and behavioral analysis. Security analysts are required to manually investigate alerts, which increases response time and operational workload. This approach becomes inefficient when dealing with sophisticated cyberattacks or large enterprise environments.

3.2 Limitations of Existing Systems

Existing System	Limitation
Manual Log Analysis	Time-consuming investigation
Static Alert Rules	High false positives
Multiple Security Tools	No centralized monitoring
Manual Incident Response	Delayed remediation
Limited Threat Intelligence	Incomplete attack context
Poor Scalability	Difficult to manage enterprise environments
Lack of AI Automation	Increased analyst workload

3.3 Summary

The limitations of existing SIEM platforms highlight the need for an intelligent security solution capable of real-time event correlation, automated incident response, behavioral analytics, and centralized security monitoring. These challenges form the foundation for the development of the proposed AegisAI Enterprise SIEM platform.

PROPOSED SYSTEM

4.1 Proposed Solution

The proposed solution, **AegisAI Enterprise SIEM – Autonomous Agentic SIEM & Incident Response Platform**, is designed to provide centralized security monitoring, intelligent threat detection, automated incident response, and real-time visualization of security events. Unlike traditional SIEM solutions, AegisAI combines endpoint monitoring, behavioral analytics, MITRE ATT&CK mapping, and AI-assisted decision support into a single integrated platform.

The platform continuously collects security telemetry from endpoints, correlates events from multiple sources, detects suspicious activities, and presents actionable insights through an interactive dashboard. Security analysts can investigate alerts, monitor live events, and initiate response actions from a centralized interface.

4.2 System Design Approach

AegisAI Enterprise SIEM follows a modular and scalable architecture designed to simplify Security Operations Center (SOC) activities. The platform collects security telemetry from multiple endpoints, processes the incoming events through intelligent detection mechanisms, correlates related security events, and presents actionable insights through a centralized dashboard. The architecture supports continuous monitoring, behavioral analysis, threat intelligence integration, and automated incident response while maintaining high performance and extensibility.

4.3 Core Functional Components

Module	Description
Dashboard	Displays real-time security metrics and system health.
Alert Center	Monitors and prioritizes detected security alerts.
AI Incident Analysis	Performs intelligent threat analysis and event correlation.
MITRE ATT&CK Mapping	Maps detected activities to MITRE ATT&CK techniques.
Threat Intelligence	Integrates external threat intelligence feeds.
Incident Timeline	Presents chronological investigation events.
Security Analytics	Generates visual reports and security statistics.
Audit Reports	Produces detailed investigation and compliance reports.

4.4 Major Advantages

- Centralized Security Monitoring
- AI-Assisted Threat Detection
- Faster Incident Response
- Interactive SOC Dashboard
- MITRE ATT&CK Integration
- Automated Security Analytics
- Enterprise-Grade Modular Architecture
- Improved Security Visibility
- Reduced Analyst Workload
- Scalable System Design

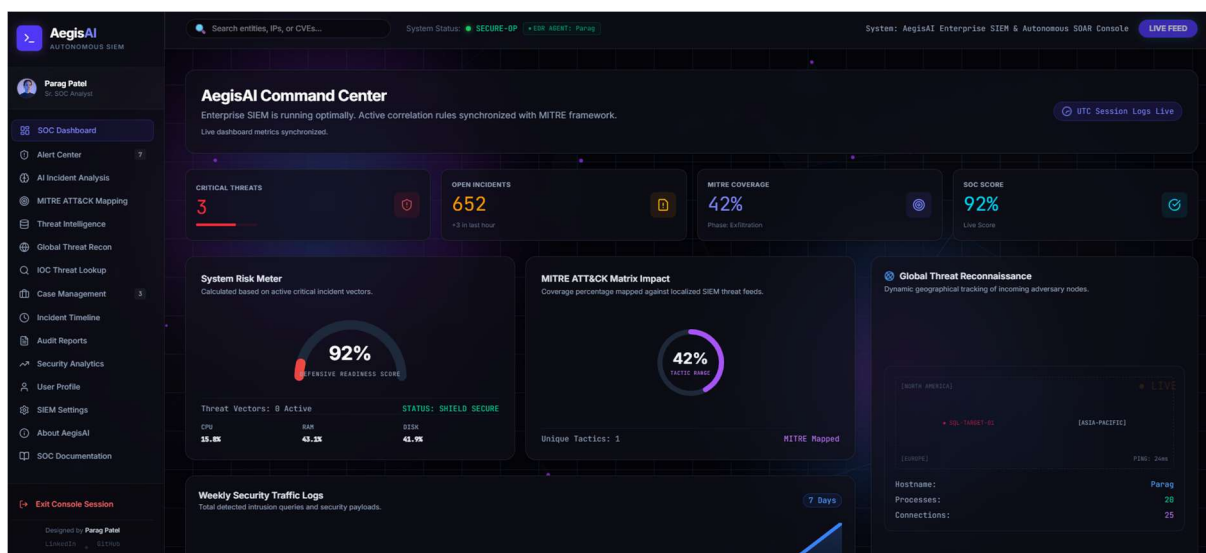


Figure 2: AegisAI Enterprise SIEM Dashboard

The dashboard provides centralized visibility of security events, critical threats, incident status, MITRE ATT&CK coverage, system health, and real-time security analytics.

PROJECT OBJECTIVES

5.1 Primary Objective

The primary objective of AegisAI Enterprise SIEM is to develop an intelligent and centralized Security Information and Event Management platform capable of continuously monitoring enterprise environments, detecting cyber threats in real time, assisting security analysts through AI-powered event correlation, and improving incident response efficiency within a modern Security Operations Center (SOC).

5.2 Specific Objectives

- ✓ Develop a centralized enterprise SIEM platform.
- ✓ Collect and monitor endpoint security telemetry.
- ✓ Detect malicious activities using intelligent correlation.
- ✓ Provide AI-assisted incident analysis.
- ✓ Integrate MITRE ATT&CK framework for attack mapping.
- ✓ Generate automated security alerts.
- ✓ Support real-time incident investigation.
- ✓ Improve visibility across enterprise infrastructure.
- ✓ Reduce manual investigation efforts.
- ✓ Provide security analytics and reporting.

5.3 Expected Outcomes

Objective	Expected Outcome
Centralized Monitoring	Improved visibility of enterprise assets
Real-Time Detection	Faster identification of cyber threats
AI Incident Analysis	Reduced analyst workload
MITRE Mapping	Better attack understanding
Automated Alerts	Faster response time

Security Analytics	Better decision making
Reporting	Improved compliance and auditing

TECHNOLOGY STACK

6.1 Technology Overview

AegisAI Enterprise SIEM is developed using a modern full-stack technology stack designed for scalability, real-time monitoring, and enterprise security operations. The platform combines frontend technologies, backend services, SIEM tools, endpoint monitoring solutions, and cybersecurity frameworks to provide centralized threat detection and incident response capabilities.

6.2 Technology Stack Table

Category	Technology	Purpose	Version
Frontend	React.js	User Interface Development	Latest
Language	TypeScript	Type Safety	Latest
Styling	Tailwind CSS	Modern UI Design	Latest
Build Tool	Vite	Frontend Build System	Latest
Backend	Node.js	Backend Runtime	Latest
Framework	Express.js	REST API Development	Latest
SIEM	Splunk Enterprise	Log Management & Analysis	Enterprise
Endpoint Security	Wazuh	Endpoint Detection & Monitoring	Latest
System Monitoring	Sysmon	Windows Event Collection	Latest
Operating System	Windows 11	Endpoint Environment	11
Threat Framework	MITRE ATT&CK	Attack Mapping	Enterprise
Version Control	Git & GitHub	Source Code Management	Latest

6.3 Core Technologies

- React.js for interactive dashboard development.
- TypeScript for reliable application development.
- Node.js and Express.js for backend services.
- Splunk Enterprise for centralized SIEM operations.
- Wazuh for endpoint monitoring and event collection.
- Sysmon for Windows system telemetry.
- MITRE ATT&CK Framework for attack mapping.
- GitHub for version control and collaboration.

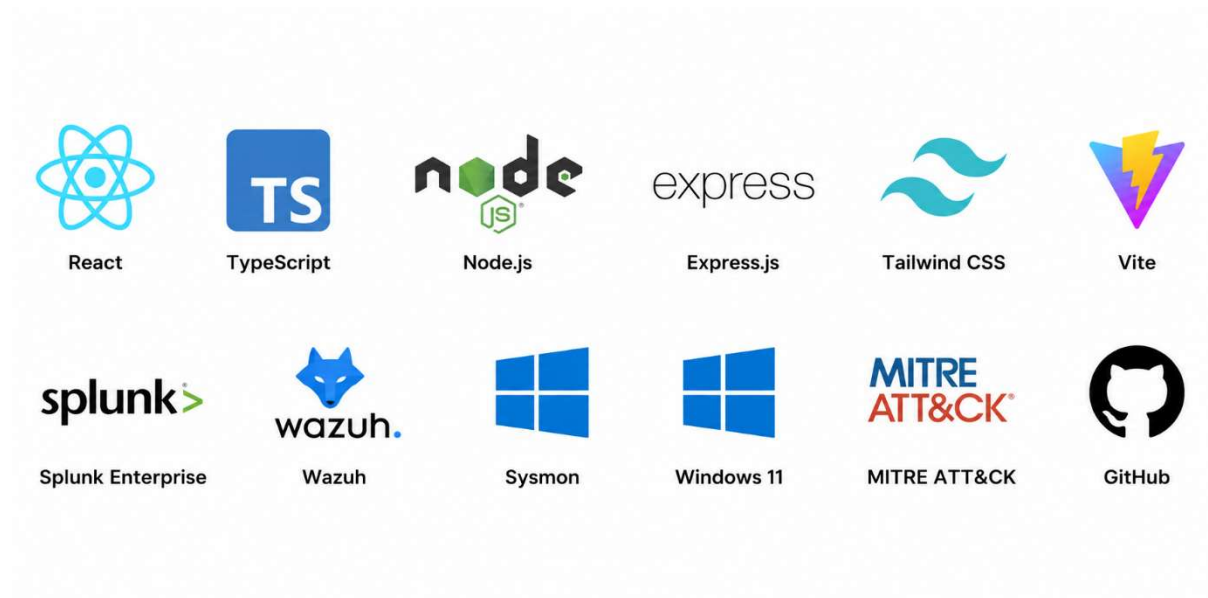


Figure 3: Technologies Used in AegisAI Enterprise SIEM

SYSTEM ARCHITECTURE

7.1 Architecture Overview

AegisAI Enterprise SIEM follows a layered architecture that enables secure data collection, centralized log processing, intelligent threat detection, and automated incident response. The architecture is designed to ensure scalability, modularity, and real-time security monitoring across enterprise environments. Each layer performs a dedicated function while communicating securely with the next layer, resulting in an efficient Security Operations Center (SOC) workflow.

7.2 System Architecture Diagram

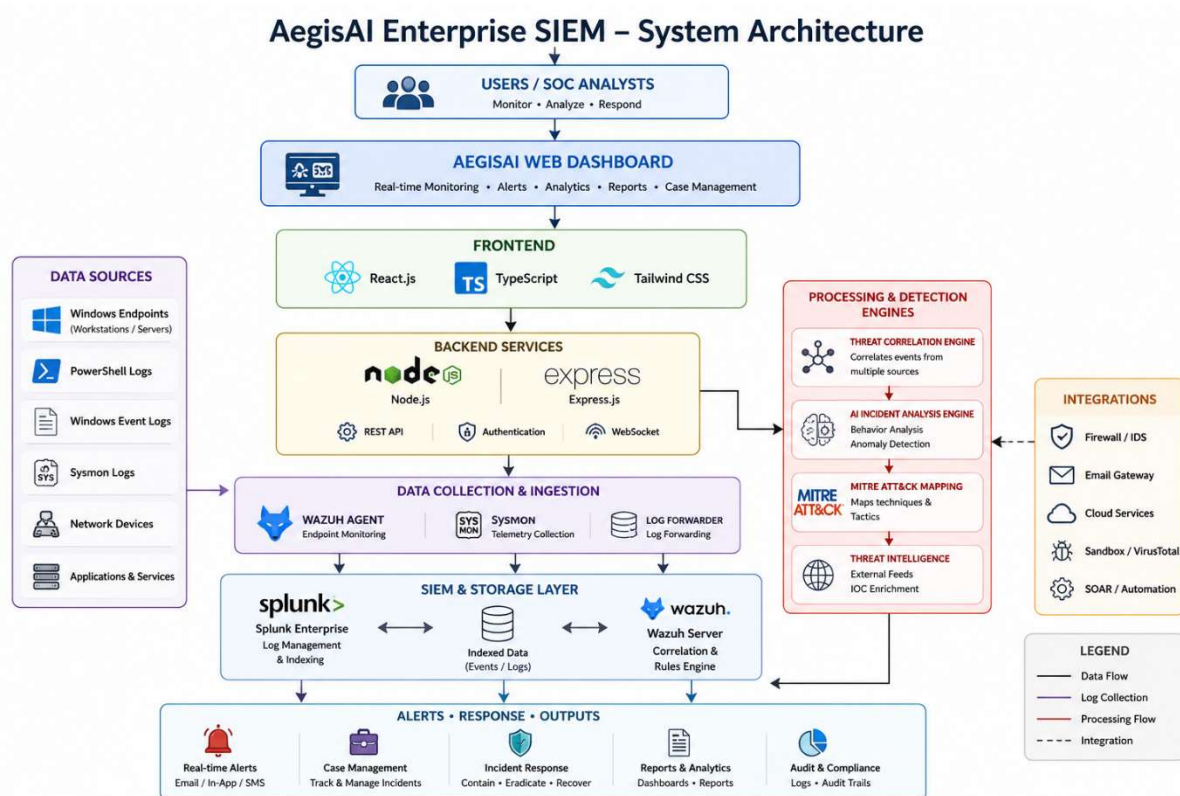


Figure 4: Overall Architecture of AegisAI Enterprise SIEM

7.3 Architecture Components

Component	Responsibility
React Dashboard	User Interface
Express API	Backend Services
Splunk Enterprise	Log Collection
Wazuh	Endpoint Monitoring
Sysmon	Windows Event Collection
AI Engine	Threat Correlation
MITRE ATT&CK	Attack Classification
Analytics Module	Reporting & Visualization

SYSTEM WORKFLOW

8.1 Workflow Overview

The AegisAI Enterprise SIEM platform follows a structured workflow that enables continuous security monitoring, intelligent event correlation, automated threat analysis, and rapid incident response. The workflow begins with collecting telemetry from multiple enterprise assets and concludes with actionable security intelligence presented through the centralized dashboard. This approach reduces manual investigation time while improving detection accuracy and operational efficiency.

8.2 Operational Workflow

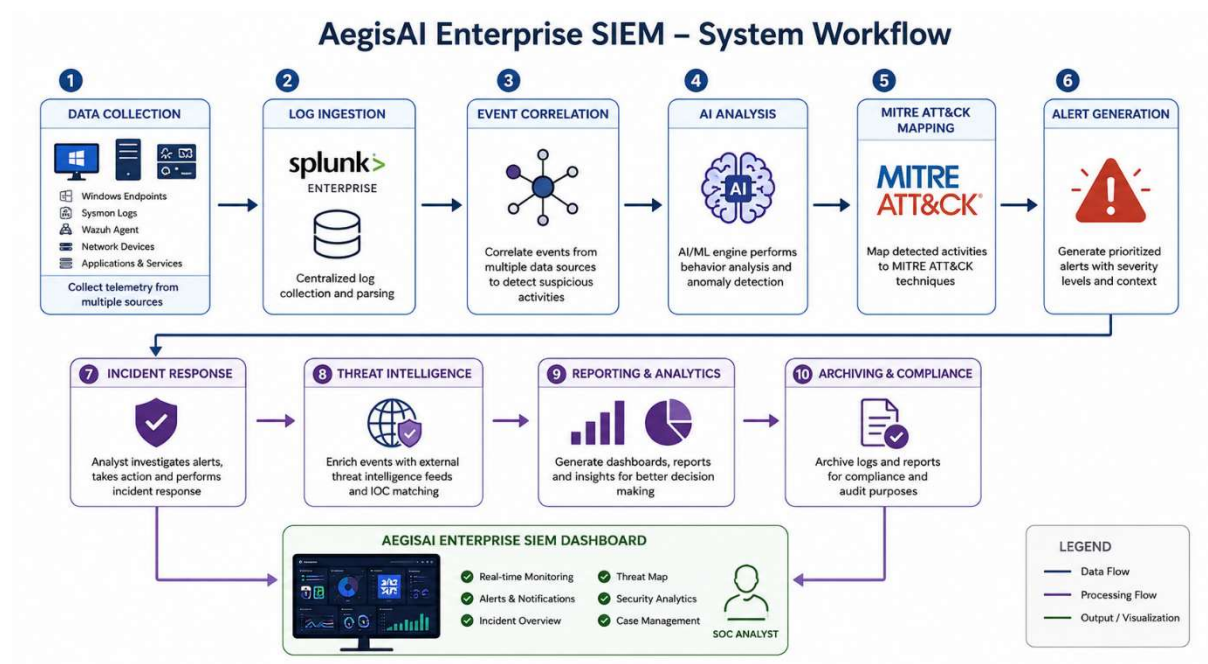


Figure 5: Operational Workflow of AegisAI Enterprise SIEM

8.3 Workflow Stages

Stage	Description
Data Collection	Collect logs from Windows, Sysmon, Wazuh and network sources.
Log Ingestion	Forward events to Splunk Enterprise for centralized processing.
Event Correlation	Identify relationships between security events.
AI Analysis	Detect anomalies using intelligent behavioral analysis.
MITRE Mapping	Associate attacks with MITRE ATT&CK techniques.
Alert Generation	Generate prioritized security alerts.
Incident Response	Support investigation and remediation workflows.
Reporting	Produce dashboards, reports and compliance records.

PROJECT MODULES

9.1 Module Overview

AegisAI Enterprise SIEM is designed using a modular architecture where each component performs a dedicated security function. This approach improves maintainability, scalability, and operational efficiency while allowing security analysts to monitor, investigate, and respond to cyber threats from a centralized platform.

9.2 Functional Modules

Module	Purpose	Status
SOC Dashboard	Centralized monitoring dashboard	Implemented
Alert Center	Displays security alerts	Implemented
AI Incident Analysis	Intelligent threat analysis	Implemented
MITRE ATT&CK Mapping	Attack technique mapping	Implemented
Threat Intelligence	External IOC integration	Implemented
Global Threat Recon	Global threat visualization	Implemented
IOC Threat Lookup	IOC search & validation	Implemented
Case Management	Incident tracking	Implemented
Incident Timeline	Event chronology	Implemented
Audit Reports	Compliance reporting	Implemented
Security Analytics	Security metrics & charts	Implemented
SIEM Settings	System configuration	Implemented

9.3 Module Summary

Each module has been designed to address a specific phase of the Security Operations Center (SOC) workflow. Together, these modules provide comprehensive visibility into enterprise security events, enabling analysts to detect, investigate, correlate, and respond to cyber incidents efficiently through a single unified platform.

SOC DASHBOARD

10.1 Dashboard Overview

The SOC Dashboard serves as the central command interface of the AegisAI Enterprise SIEM platform. It provides security analysts with a unified view of the organization's security posture by presenting real-time alerts, system health, incident statistics, MITRE ATT&CK coverage, and operational metrics. The dashboard enables analysts to quickly identify high-priority threats and make informed decisions without switching between multiple security tools.

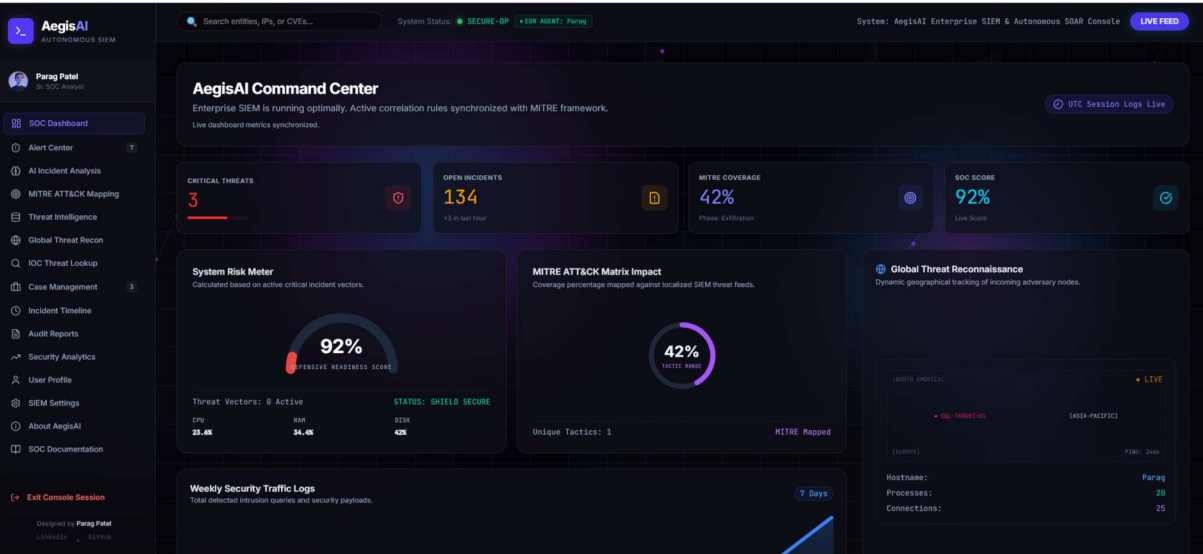


Figure 6: AegisAI Enterprise SIEM – Main SOC Dashboard

10.2 Dashboard Components

Component	Description
System Status	Displays overall platform health.
Critical Threats	Shows high-severity detected threats.
Open Incidents	Displays active investigation cases.
MITRE Coverage	Indicates mapped ATT&CK techniques.
SOC Score	Represents overall security posture.

System Risk Meter	Measures endpoint security readiness.
Threat Recon	Displays global threat intelligence.
Navigation Panel	Access to all security modules.

10.3 Key Benefits

- Centralized Security Visibility
- Real-Time Threat Monitoring
- Fast Incident Investigation
- Interactive Security Dashboard
- Improved Analyst Productivity
- Better Decision Making
- Enterprise-Grade User Experience

ALERT CENTER

11.1 Module Overview

The Alert Center is one of the core operational modules of the AegisAI Enterprise SIEM platform. It provides a centralized interface for monitoring, prioritizing, and managing security alerts generated from multiple security sources. Each alert contains detailed information regarding severity, affected assets, event timestamps, operational status, and recommended response actions, enabling analysts to perform efficient incident triage and investigation.

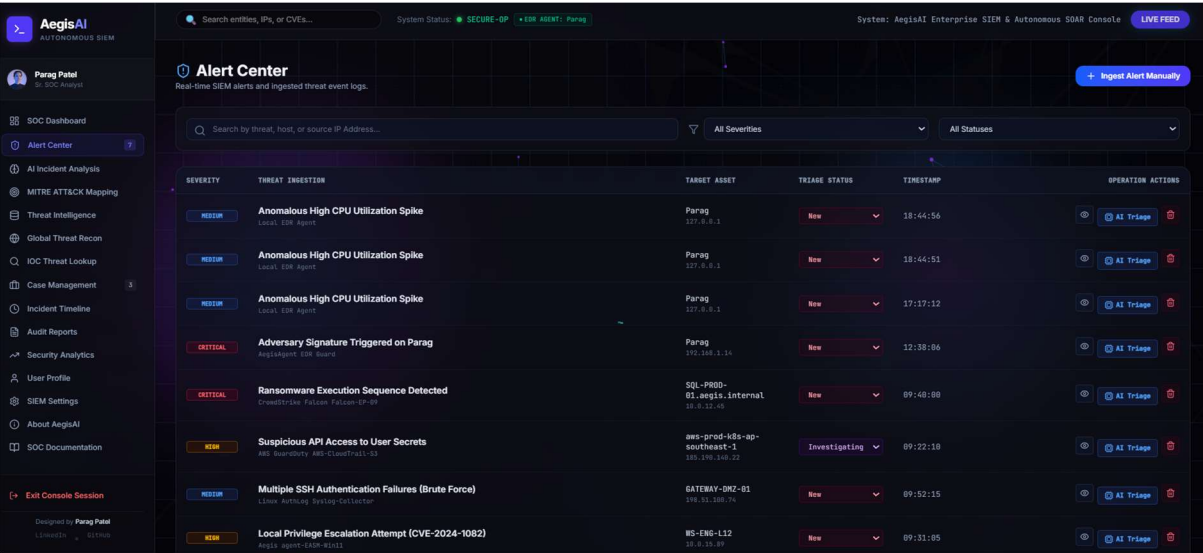


Figure 7: Alert Center displaying real-time security alerts and incident triage.

11.2 Key Features

Feature	Description
Real-Time Alerts	Displays live security events generated across monitored systems.
Severity Classification	Categorizes alerts into Critical, High, Medium and Low severity levels.
Advanced Search	Search alerts using hostnames, IP addresses, CVEs and event names.
Alert Filtering	Filter alerts based on severity and operational status.
AI Triage	AI-assisted prioritization and preliminary incident analysis.
Asset Information	Displays affected endpoint and host information.

Status Tracking	Tracks alert states such as New, Investigating and Resolved.
Analyst Actions	View, investigate or remove alerts directly from the dashboard.

11.3 Benefits

- Real-time security monitoring
- Centralized alert management
- Faster incident prioritization
- AI-assisted investigation
- Improved SOC analyst productivity
- Reduced response time
- Better visibility into enterprise threats

11.4 Summary

The Alert Center significantly enhances Security Operations Center (SOC) efficiency by consolidating security events into a single interface. With intelligent filtering, severity-based prioritization, and AI-assisted triage, analysts can rapidly identify high-risk incidents and initiate appropriate response actions, reducing both investigation time and operational complexity.

AI INCIDENT ANALYSIS

12.1 Module Overview

The AI Incident Analysis module enhances the traditional Security Operations Center (SOC) workflow by incorporating intelligent threat analysis, automated event correlation, and AI-assisted decision support. Instead of relying solely on manual investigation, the system analyzes security events, identifies attack patterns, evaluates risk severity, and recommends appropriate remediation actions. This significantly reduces investigation time while improving the accuracy and consistency of incident handling.

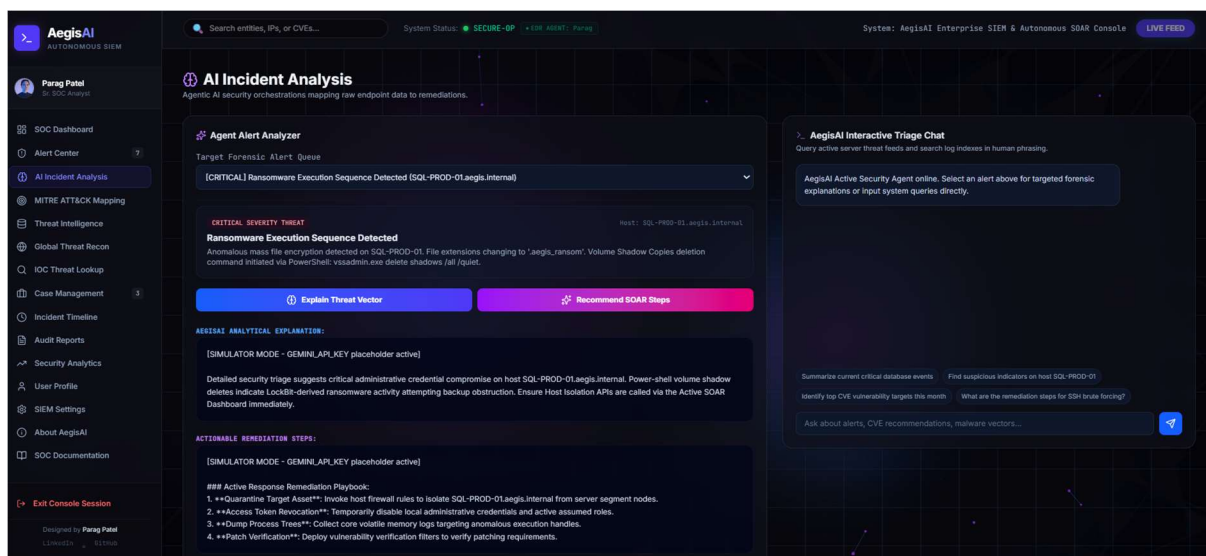


Figure 8: AI Incident Analysis Module with Intelligent Threat Investigation and SOAR Recommendations

12.2 Functional Components

Component	Function
Agent Alert Analyzer	Processes incoming security alerts.
Threat Explanation	Generates AI-based explanations of detected attacks.
Risk Assessment	Evaluates threat severity and impact.
SOAR Recommendations	Suggests automated incident response actions.
Interactive AI Chat	Allows analysts to query security events using natural language.
Behavioral Analysis	Detects suspicious activity using intelligent correlation.

Active Remediation	Provides step-by-step response recommendations.
Analyst Assistance	Accelerates investigation and decision-making.

12.3 AI Analysis Workflow

- Collect security alerts from SIEM.
- Analyze attack indicators using AI.
- Correlate related security events.
- Generate detailed threat explanation.
- Recommend SOAR remediation steps.
- Assist analyst through interactive AI chat.
- Support rapid incident investigation.
- Reduce Mean Time To Respond (MTTR).

12.4 Benefits

- AI-Assisted Threat Investigation
- Intelligent Event Correlation
- Faster Incident Response
- Reduced Manual Analysis
- Automated Remediation Guidance
- Interactive Security Assistant
- Improved SOC Productivity
- Better Decision Making

12.5 Summary

The AI Incident Analysis module transforms traditional security operations by integrating artificial intelligence into incident investigation. Through automated threat explanation, intelligent event correlation, and contextual remediation guidance, the platform enables SOC analysts to identify, understand, and respond to cyber threats more efficiently than conventional manual approaches.

MITRE ATT&CK MAPPING

13.1 Module Overview

The MITRE ATT&CK Mapping module enables security analysts to visualize detected attack techniques based on the globally recognized MITRE ATT&CK framework. Security events are automatically correlated with ATT&CK tactics and techniques, allowing analysts to understand attacker behavior, identify affected phases of the cyber kill chain, and prioritize appropriate defensive actions. This mapping improves threat visibility and supports faster incident investigation.

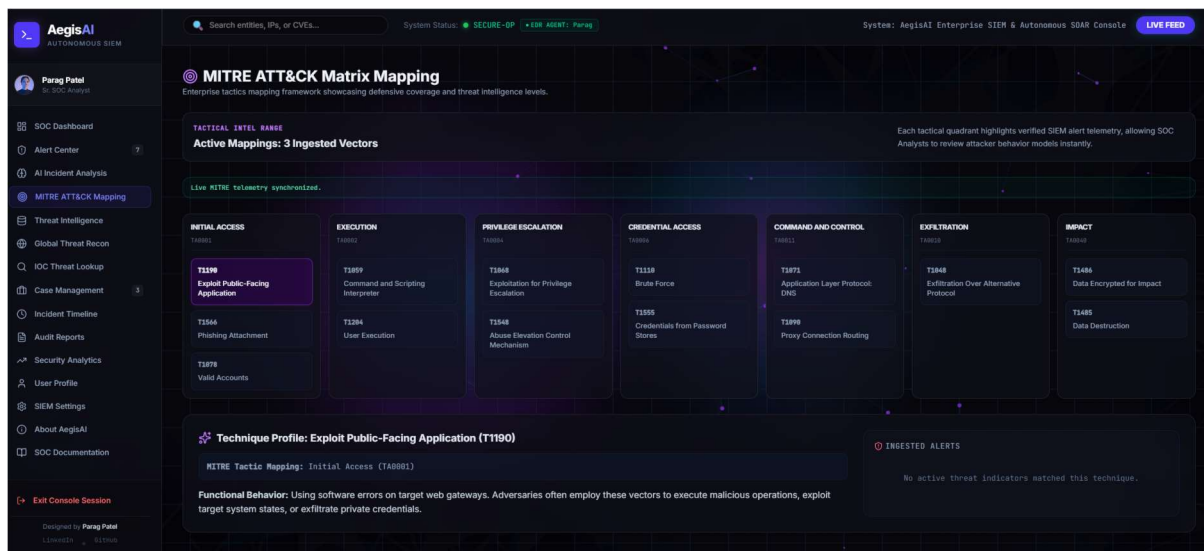


Figure 9: MITRE ATT&CK Matrix Mapping Dashboard

13.2 Supported ATT&CK Tactics

MITRE Tactic	Description
Initial Access	Entry point used by attackers to gain access.
Execution	Malicious code execution techniques.
Privilege Escalation	Gaining elevated system privileges.
Credential Access	Theft or abuse of authentication credentials.
Command and Control	Remote communication with compromised systems.

Exfiltration	Unauthorized extraction of sensitive information.
Impact	Activities intended to disrupt, encrypt or destroy systems.

13.3 Module Features

- Automatic MITRE ATT&CK Mapping
- Enterprise Attack Visualization
- Tactic & Technique Identification
- Interactive Threat Navigation
- ATT&CK Technique Profiles
- Threat Correlation Engine
- Security Coverage Analysis
- SOC Investigation Support

13.4 Benefits

Benefit	Business Value
Faster Threat Investigation	Reduces analyst investigation time by providing contextual attack mapping.
Standardized Framework	Uses the globally accepted MITRE ATT&CK knowledge base for threat classification.

13.5 Summary

The MITRE ATT&CK Mapping module strengthens threat investigation by transforming raw security alerts into structured attack intelligence. By correlating events with MITRE tactics and techniques, the platform enables SOC analysts to understand attacker behavior, identify affected attack phases, and improve incident response using an industry-standard framework.

THREAT INTELLIGENCE

14.1 Module Overview

The Threat Intelligence module enriches security operations by collecting, storing, and analyzing Indicators of Compromise (IOCs) from internal and external intelligence sources. It enables security analysts to identify malicious IP addresses, domains, malware families, threat actors, and emerging attack patterns. By correlating intelligence feeds with real-time security events, the platform improves threat visibility, accelerates investigations, and enhances proactive defense capabilities.

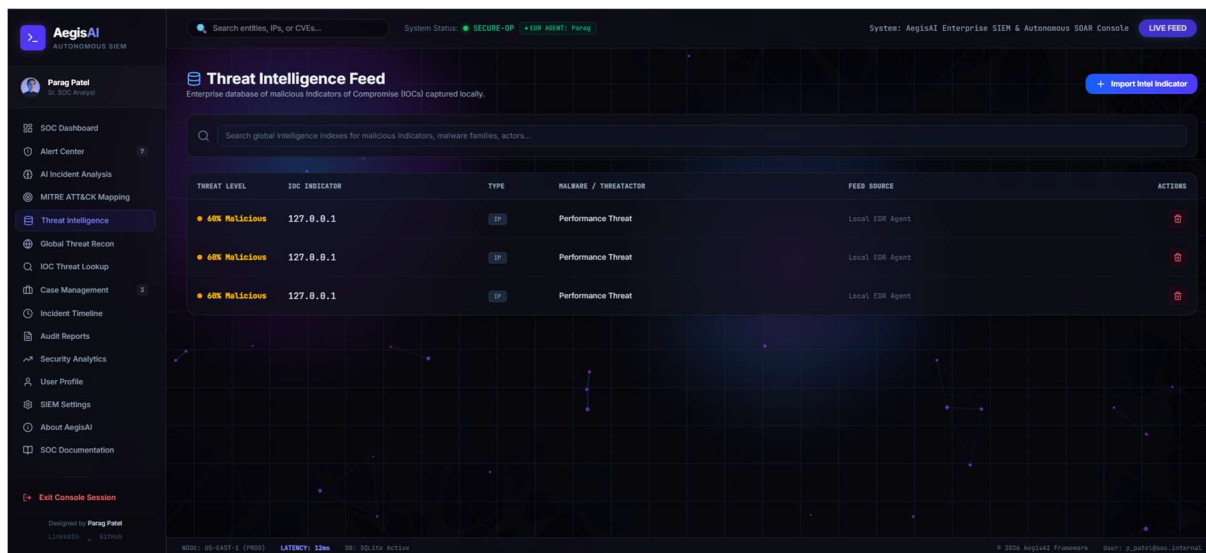


Figure 10: Threat Intelligence Feed displaying Indicators of Compromise (IOCs)

14.2 Threat Intelligence Features

Feature	Description
IOC Database	Stores malicious IPs, domains and hashes.
Threat Feed Integration	Imports intelligence from trusted sources.
Malware Identification	Detects known malware families.
Threat Actor Mapping	Associates attacks with threat groups.
IOC Search	Search indicators using IPs, domains or hashes.

Threat Classification	Classifies threats based on severity.
Feed Management	Manages imported intelligence feeds.
IOC Correlation	Matches IOCs against collected security events.

14.3 Advantages

- Centralized Threat Intelligence Repository
- Faster IOC Investigation
- Improved Threat Hunting
- Better Detection Accuracy
- Early Threat Identification
- Reduced False Positives
- Enhanced Security Visibility
- Enterprise Threat Correlation

14.4 Summary

The Threat Intelligence module strengthens enterprise cyber defense by providing contextual intelligence for detected security events. Through IOC correlation, malware identification, and threat feed integration, the platform enables analysts to identify potential threats quickly and improve the overall effectiveness of security investigations.

GLOBAL THREAT RECONNAISSANCE

15.1 Module Overview

The Global Threat Reconnaissance module enables analysts to perform real-time reconnaissance on domains, IP addresses, and network assets using OSINT (Open Source Intelligence) techniques. It combines DNS resolution, Geo-IP analysis, ASN information, network telemetry, and live reconnaissance logs to provide comprehensive visibility into external attack surfaces and potential threats. This module assists analysts in identifying suspicious infrastructure and gathering contextual intelligence before initiating incident response.

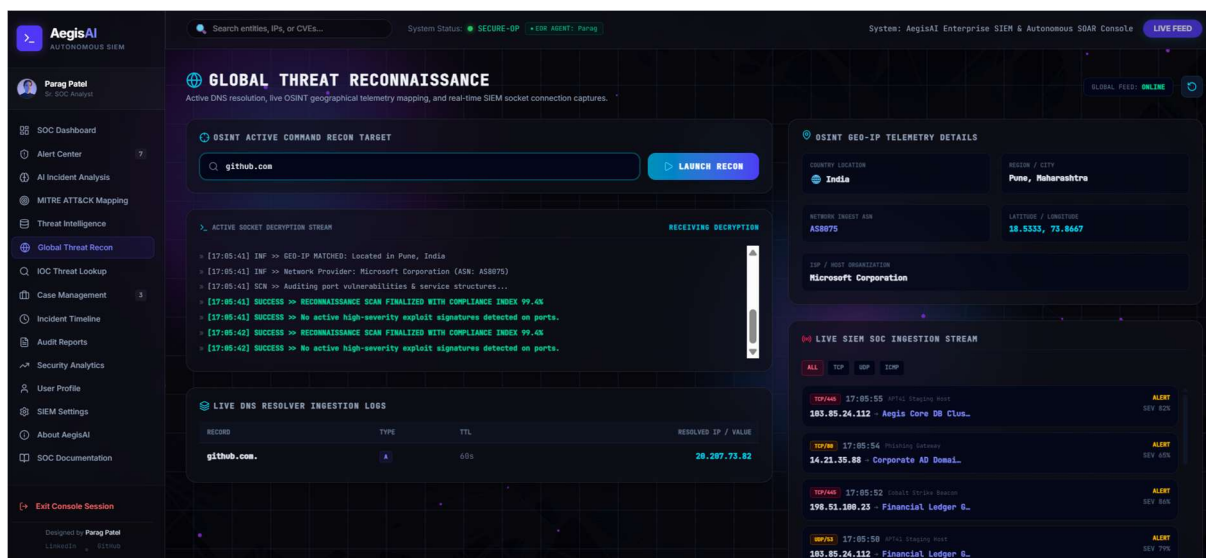


Figure 11: Global Threat Reconnaissance Module with OSINT, Geo-IP Analysis and Live SIEM Monitoring

15.2 Key Functionalities

Function	Description
Domain Reconnaissance	Analyze target domains and network assets.
DNS Resolution	Resolve domain names to IP addresses.
Geo-IP Intelligence	Identify country, city and geographic location.
ASN Information	Display Autonomous System Network details.
Live Telemetry	Monitor real-time reconnaissance events.

SIEM Stream	Display live network activity and security events.
OSINT Integration	Enrich investigations with open-source intelligence.
Threat Validation	Verify reconnaissance results before response actions.

15.3 Advantages

- Real-Time Domain Investigation
- Integrated OSINT Intelligence
- Geo-IP Visualization
- Network Attribution
- Faster Threat Hunting
- External Attack Surface Visibility
- Enhanced Reconnaissance Capabilities
- Improved Security Awareness

15.4 Summary

The Global Threat Reconnaissance module extends the capabilities of the SIEM platform beyond internal monitoring by providing external intelligence and reconnaissance features. Through DNS analysis, geographic attribution, network telemetry, and OSINT enrichment, analysts can gain valuable context about suspicious infrastructure and improve the effectiveness of cyber threat investigations.

IOC THREAT LOOKUP

16.1 Module Overview

The IOC (Indicators of Compromise) Threat Lookup module enables security analysts to validate suspicious IP addresses, domains, URLs, and file hashes against enterprise threat intelligence repositories. The module performs rapid indicator correlation, provides contextual threat information, and assists analysts in determining the reputation and risk level of suspected indicators. This capability significantly improves threat hunting, malware investigation, and incident response workflows.

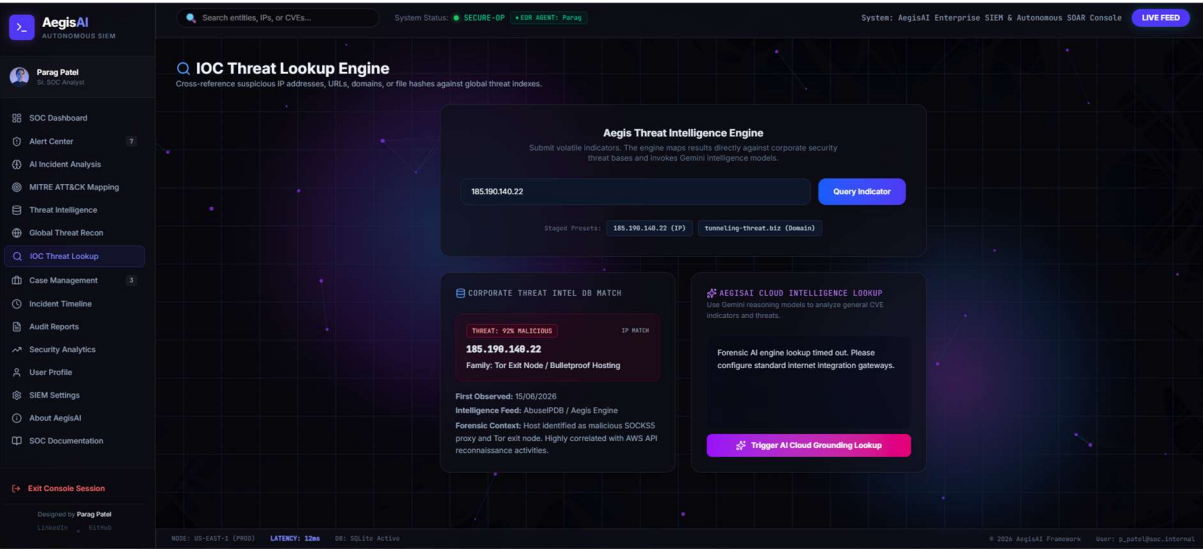


Figure 12: IOC Threat Lookup Engine with Enterprise Threat Intelligence Correlation

16.2 Key Features

Feature	Description
IOC Search Engine	Search suspicious IPs, domains, URLs and hashes.
Threat Intelligence Database	Matches indicators against enterprise intelligence feeds.
Risk Classification	Classifies indicators by maliciousness score.
Threat Reputation	Displays confidence level and threat family.
AI Threat Lookup	Uses AI-assisted analysis for indicator investigation.

Contextual Intelligence	Provides historical forensic information.
Threat Correlation	Correlates IOCs with security events.
Investigation Support	Assists SOC analysts during threat hunting.
Enterprise Intelligence	Integrates corporate and external threat feeds.

16.3 Investigation Workflow

- Submit suspicious IOC.
- Search enterprise threat database.
- Match intelligence feeds.
- Evaluate malicious score.
- Display forensic context.
- Correlate with existing incidents.
- Generate investigation result.
- Assist SOC analyst.

16.4 Benefits

- Faster IOC Investigation
- AI-Assisted Threat Lookup
- Enterprise Threat Intelligence
- Reduced Investigation Time
- Improved Threat Hunting
- Better Malware Identification
- Centralized IOC Repository

16.5 Summary

The IOC Threat Lookup module enables rapid validation of suspicious indicators using enterprise threat intelligence and AI-assisted investigation techniques. By providing contextual intelligence, maliciousness scoring, and historical forensic information, the module helps SOC analysts make faster and more accurate security decisions during cyber incident investigations.

CASE MANAGEMENT

17.1 Module Overview

The Case Management module provides a centralized environment for tracking, assigning, investigating, and documenting cybersecurity incidents throughout their lifecycle. It enables SOC analysts to maintain structured investigation records, collaborate on incidents, preserve forensic evidence, and ensure accountability during security operations. By integrating incident details, investigation logs, analyst assignments, and digital forensic tools, the module streamlines the complete incident response process.

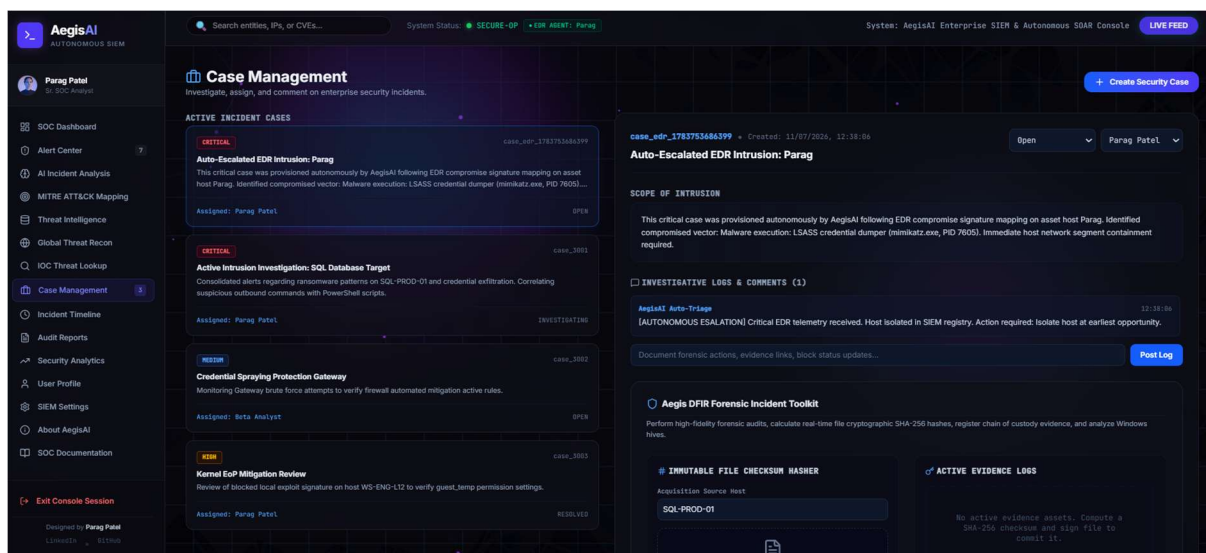


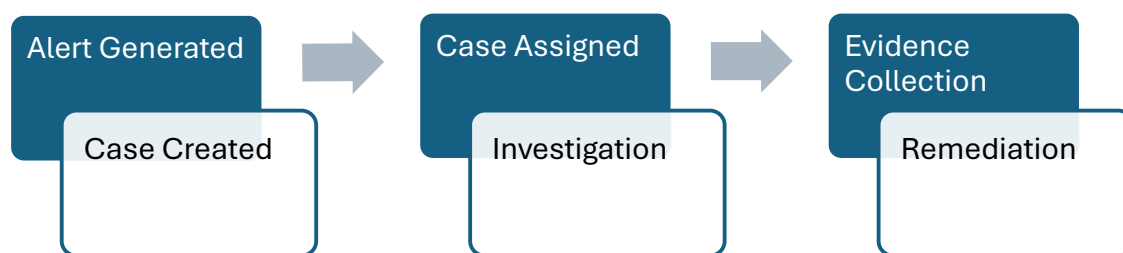
Figure 13: Case Management Module with Incident Tracking and Digital Forensics

17.2 Functional Components

Component	Description
Incident Cases	Displays all active and historical security cases.
Case Assignment	Assigns incidents to SOC analysts.
Case Status	Tracks Open, Investigating, and Resolved cases.
Investigation Logs	Records analyst comments and forensic observations.
Incident Scope	Documents attack details and affect assets.
Evidence Management	Maintains digital evidence and chain of custody.

DFIR Toolkit	Supports forensic acquisition and integrity verification.
SHA-256 Hash Verification	Validates evidence integrity using cryptographic hashing.
Analyst Collaboration	Enables team-based investigation and response.

17.3 Incident Response Lifecycle



17.4 Key Benefits

- Centralized Incident Management
- Faster Case Assignment
- Digital Forensics Support
- Secure Evidence Handling
- Improved Team Collaboration
- Complete Investigation Audit Trail
- Reduced Incident Resolution Time
- Enterprise Incident Tracking

17.5 Summary

The Case Management module serves as the operational backbone of the incident response workflow. By integrating incident documentation, forensic evidence management, analyst collaboration, and structured case tracking, the platform ensures that every security incident is investigated, documented, and resolved in a systematic and auditable manner.

INCIDENT TIMELINE

18.1 Module Overview

The Incident Timeline module provides a chronological view of all security events associated with a specific incident. It records every stage of the investigation, including alert generation, automated analysis, analyst actions, containment activities, and remediation steps. By maintaining a structured timeline, the platform enables security analysts to reconstruct attack sequences, verify response actions, and maintain a complete audit trail for forensic investigations.

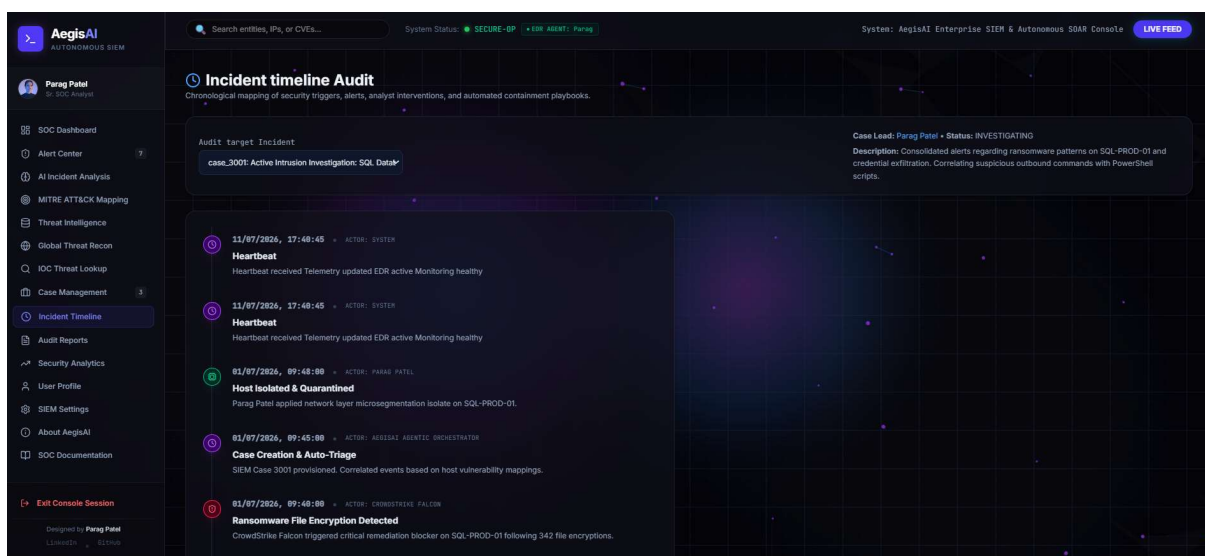


Figure 14: Incident Timeline showing chronological security investigation events.

18.2 Timeline Events

Timeline Event	Description
Alert Generated	Initial security event detected by SIEM.
Auto-Triage	AI engine correlates related security events.
Case Creation	Incident automatically registered.
Analyst Assignment	Case assigned to SOC analyst.
Host Isolation	Compromised endpoint quarantined.
Investigation	Manual forensic investigation performed.
Remediation	Threat removed and system secured.

Case Closure	Incident resolved and documented.
--------------	-----------------------------------

18.3 Timeline Features

- Chronological Event Tracking
- AI & Analyst Activity Logging
- Automated Containment History
- Complete Investigation Audit Trail
- Time-Stamped Security Events
- Incident Reconstruction
- Forensic Evidence Correlation
- Compliance Documentation

18.4 Benefits

Benefit	Impact
End-to-End Visibility	Complete record of every investigation step.
Faster Incident Review	Simplifies post-incident analysis and auditing.

18.5 Summary

The Incident Timeline module provides a complete chronological record of every security event throughout the incident lifecycle. This structured audit trail enables analysts to reconstruct attack sequences, validate remediation actions, and support digital forensic investigations while ensuring transparency and compliance.

AUDIT REPORTS

19.1 Module Overview

The Audit Reports module enables security teams to generate professional, compliance-ready reports for executive management, technical analysts, and incident response teams. It consolidates security findings, risk assessments, investigation summaries, and system telemetry into structured reports that support cybersecurity governance, regulatory compliance, and organizational decision-making. Reports can be exported in multiple formats for documentation, review, and long-term record keeping.

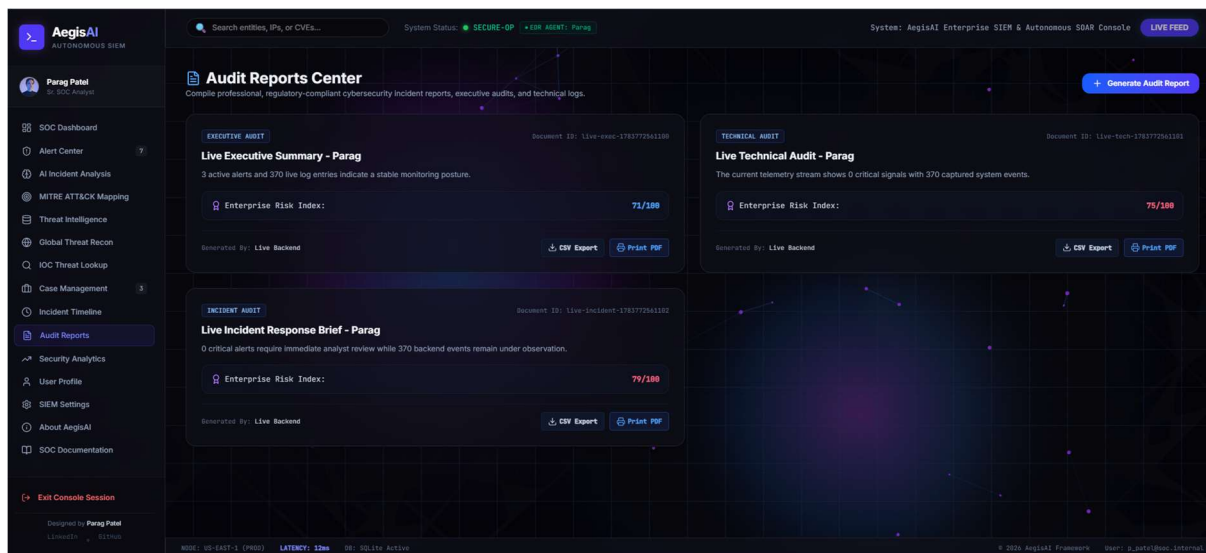


Figure 15 : Audit Reports Center displaying Executive, Technical and Incident Audit Reports

19.2 Report Categories

Report Type	Description
Executive Audit	High-level security posture and enterprise risk summary.
Technical Audit	Detailed technical findings and security telemetry analysis.
Incident Response Report	Documents incident investigations and remediation activities.
Risk Assessment Report	Evaluates organizational cyber risk exposure.
Compliance Report	Supports regulatory and security compliance requirements.

Export Reports	Generates reports in PDF and CSV formats.
----------------	---

19.3 Key Features

- Executive Security Summary
- Technical Audit Reports
- Incident Response Documentation
- Enterprise Risk Assessment
- PDF Report Export
- CSV Data Export
- Compliance Reporting
- Professional Report Generation

19.4 Benefits

Benefit	Impact
Automated Reporting	Saves time by generating standardized security reports automatically.
Compliance Support	Simplifies audits and regulatory documentation requirements.

19.5 Summary

The Audit Reports module transforms operational security data into structured documentation that can be shared with technical teams, management, and compliance auditors. Through automated report generation and export capabilities, the platform improves operational transparency, supports governance requirements, and strengthens cybersecurity documentation practices.

SECURITY ANALYTICS

20.1 Module Overview

The Security Analytics module provides real-time visualization of enterprise security metrics through interactive dashboards, performance indicators, and operational charts. It enables SOC analysts to monitor system utilization, endpoint health, security events, and overall infrastructure performance from a centralized interface. By transforming raw telemetry into meaningful visual analytics, the module supports informed decision-making, continuous monitoring, and proactive threat management.

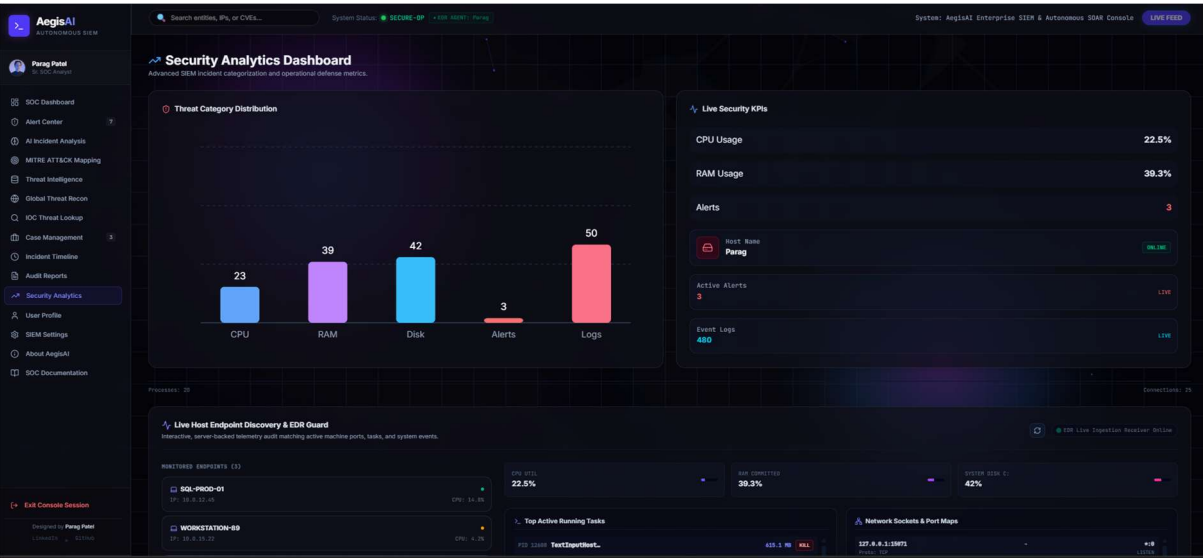


Figure 16: Security Analytics Dashboard showing system performance and operational security metrics

20.2 Dashboard Components

Component	Description
Threat Category Chart	Visualizes security event distribution.
Live Security KPIs	Displays CPU, RAM, Alerts and Event Logs.
Endpoint Monitoring	Shows monitored hosts and endpoint status.
Resource Utilization	Tracks CPU, Memory and Disk usage.
Running Processes	Displays active system processes.

Network Monitoring	Monitors sockets, ports and network activity.
Security Metrics	Provides operational security statistics.
Real-Time Dashboard	Continuously updates security information.
Performance Indicators	Measures overall platform health and efficiency.

20.3 Key Performance Indicators (KPIs)

KPI	Purpose
CPU Usage	Monitor processor utilization.
RAM Usage	Track memory consumption.
Disk Usage	Monitor storage utilization.
Active Alerts	Display current security alerts.
Event Logs	Count collected security events.
Endpoint Status	Monitor connected endpoints.

20.4 Benefits

- Interactive Security Dashboards
- Live Performance Monitoring
- Real-Time Operational Metrics
- Endpoint Health Monitoring
- Improved Resource Visibility
- Faster Decision Making
- Enhanced Security Awareness
- Enterprise Performance Reporting

20.5 Summary

The Security Analytics module transforms operational telemetry into meaningful visual insights that help SOC analysts monitor infrastructure health, evaluate security posture, and identify abnormal system behavior. Through real-time dashboards and performance metrics, the platform enhances situational awareness and supports data-driven cybersecurity operations.

USER PROFILE

21.1 Module Overview

The User Profile module provides a centralized interface for managing SOC analyst information, access privileges, professional roles, and administrative identity. It enables authorized users to maintain profile details, contact information, investigator roles, and organizational responsibilities while ensuring secure access to the Security Operations Center. This module supports identity management and accountability within enterprise cybersecurity operations.

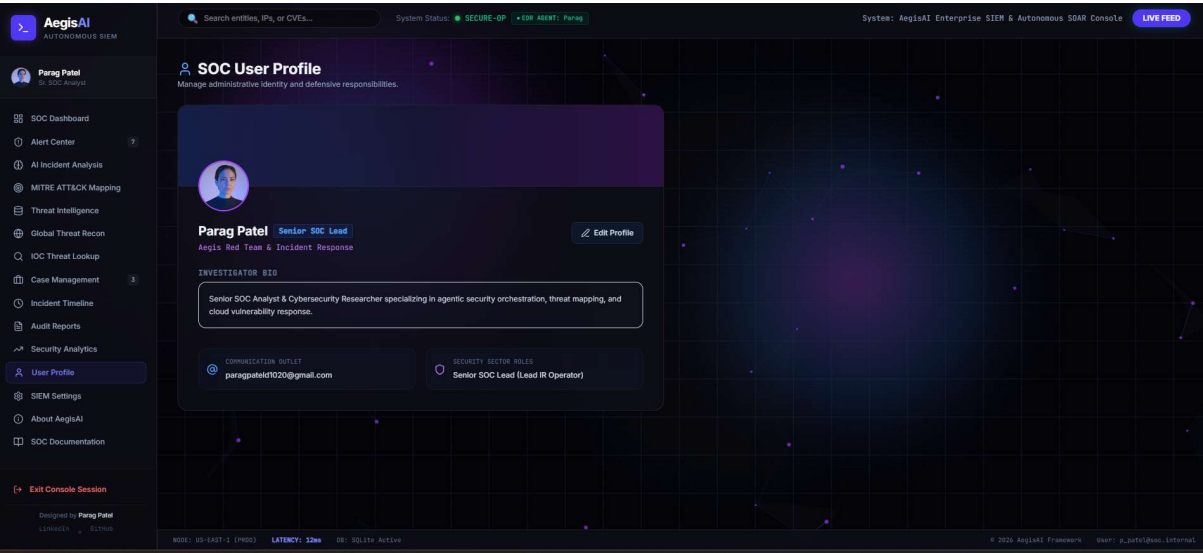


Figure 17 : SOC User Profile and Analyst Identity Management

21.2 Profile Components

Component	Description
Analyst Profile	Displays investigator information.
Role Management	Shows assigned SOC responsibilities.
Contact Details	Stores analyst communication details.
Investigator Biography	Maintains analyst profile summary.
Identity Management	Secure analyst identity records.
Access Privileges	Controls platform permissions.

Profile Editing	Allows profile updates.
Administrative Access	Supports secure account management.

21.3 Benefits

- Centralized Analyst Identity
- Role-Based Access Control
- Secure User Management
- Better Accountability
- Professional Profile Management
- Simplified Administration

21.4 Summary

The User Profile module provides a secure and organized environment for managing analyst identities, access permissions, and professional information. It improves administrative control while ensuring accountability within enterprise SOC operations.

SIEM SETTINGS

22.1 Module Overview

The SIEM Settings module enables administrators to configure system parameters, telemetry retention policies, notification preferences, cloud integrations, and automated response settings. Through a centralized configuration interface, administrators can customize platform behavior, manage third-party security integrations, and optimize security monitoring based on organizational requirements.

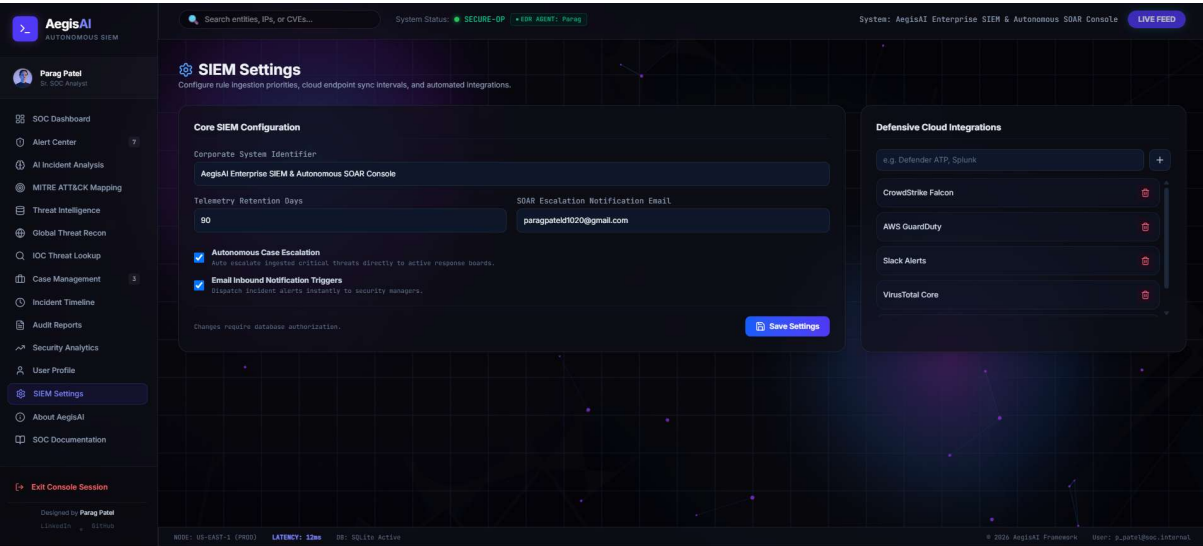


Figure 18: SIEM Configuration and Security Management Settings

22.2 Configuration Options

Configuration	Purpose
System Identifier	Defines the SIEM instance name.
Telemetry Retention	Configures log retention duration.
Email Notifications	Sends security alerts to administrators.
Autonomous Escalation	Automatically escalates critical incidents.
Cloud Integrations	Connects third-party security platforms.
Security Policies	Defines monitoring and response rules.
Notification Settings	Controls alert delivery mechanisms.

Configuration Management	Saves and updates platform settings.
--------------------------	--------------------------------------

22.3 Integrated Platforms

- CrowdStrike Falcon
- AWS GuardDuty
- Microsoft Defender
- VirusTotal
- Slack Notifications
- Splunk Enterprise
- Wazuh
- Email Alert Service

22.4 Benefits

- Flexible SIEM Configuration
- Automated Alert Management
- Third-Party Security Integration
- Centralized Administration
- Improved Operational Control
- Enterprise Security Automation

22.5 Summary

The SIEM Settings module provides centralized configuration and administration capabilities for the AegisAI Enterprise SIEM platform. By supporting customizable policies, cloud integrations, automated notifications, and security preferences, it enables organizations to tailor the platform according to operational and compliance requirements.

ABOUT AEGISAI

23.1 Module Overview

The About AegisAI module provides an overview of the AegisAI Enterprise SIEM platform, including project objectives, developer profile, AI-driven security philosophy, and core operational concepts. It introduces the vision behind the platform and highlights how artificial intelligence enhances enterprise threat detection, automated triage, and incident response. This section serves as the project identity and technical overview of the developed solution.

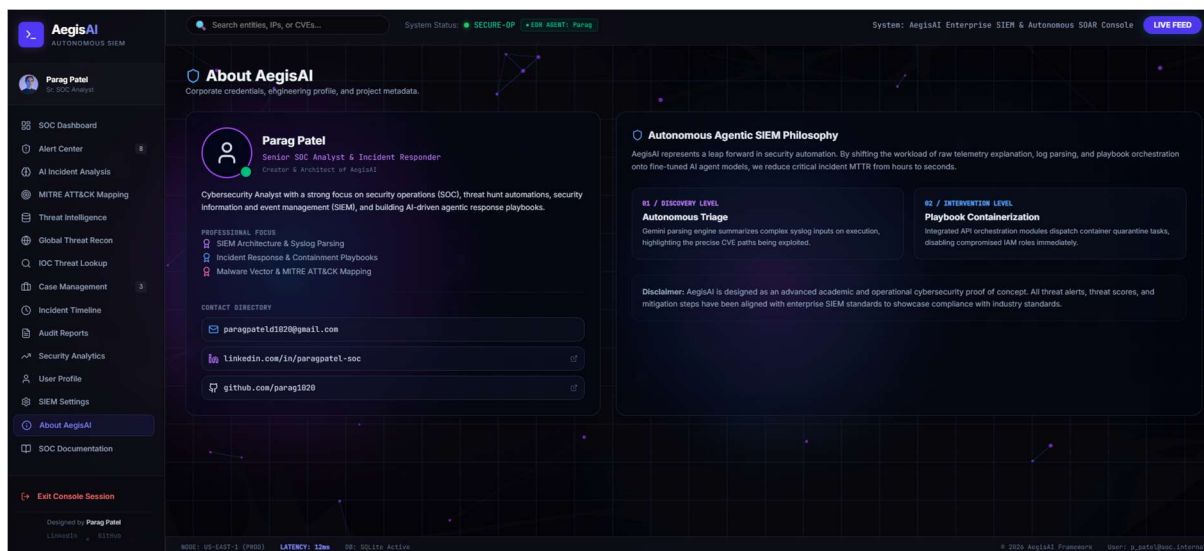


Figure 19: About AegisAI Module

23.2 Module Features

Feature	Description
Developer Profile	Displays project developer information.
AI Philosophy	Explains AI-driven SIEM architecture.
Security Objectives	Defines project goals and capabilities.
Contact Information	Professional contact details.
Project Metadata	Version and platform information.
AI Security Model	Overview of intelligent response workflow.

23.3 Key Highlights

- AI-powered Enterprise SIEM
- Autonomous Incident Response
- Threat Intelligence Integration
- Automated Playbooks
- Enterprise-ready Architecture
- Modern SOC Operations

23.4 Summary

The About AegisAI module introduces the project's architecture, objectives, and AI-powered security philosophy. It provides users with a clear understanding of the platform's purpose and demonstrates how intelligent automation improves cybersecurity operations.

SOC DOCUMENTATION

24.1 Module Overview

The SOC Documentation module serves as the technical knowledge repository of the AegisAI Enterprise SIEM platform. It contains architecture diagrams, incident response workflows, deployment procedures, operational guidelines, and analyst handbooks. This module assists security analysts in understanding system architecture, executing incident response processes, and deploying monitoring agents across different operating systems.

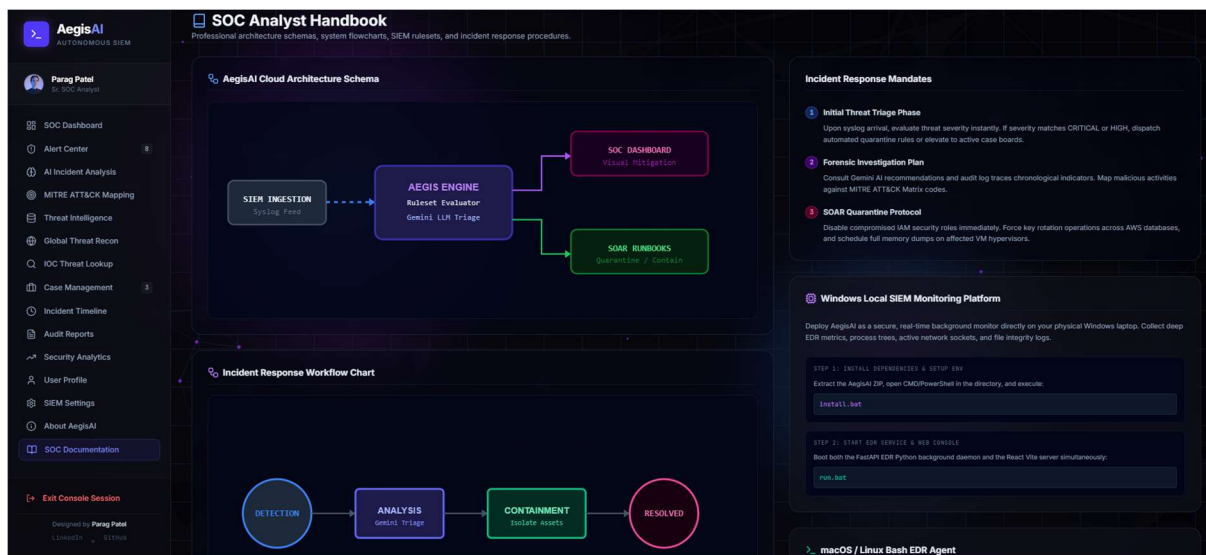


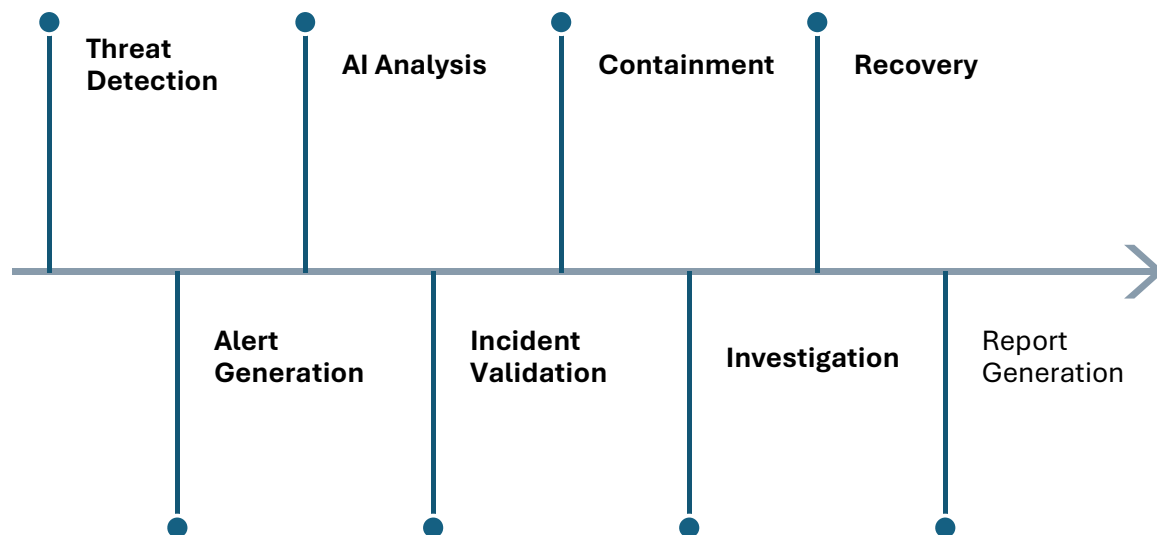
Figure 20 : SOC Documentation and Incident Response Guide

24.2 Documentation Sections

Section	Purpose
Cloud Architecture	Shows SIEM architecture flow.
Incident Workflow	Defines security response lifecycle.
Analyst Handbook	Operational procedures for SOC analysts.
Windows Deployment	Installation guide for Windows agent.
Linux Deployment	Bash-based monitoring deployment.
EDR Capabilities	Lists endpoint monitoring features.

Response Procedures	Standard incident handling process.
Technical Documentation	Platform operational reference.

24.3 Incident Response Workflow



24.4 Deployment Process

Operating System	Deployment Method
Windows	install.bat
Windows Service	run.bat
Linux	Bash Script
Endpoint	Local EDR Agent
Dashboard	React Web Interface
Backend	FastAPI Service

24.5 Benefits

- Standardized Documentation
- Easy Deployment
- Faster Analyst Training
- Structured Incident Handling

- Better Knowledge Sharing
- Improved Operational Efficiency

24.6 Summary

The SOC Documentation module centralizes all operational procedures, deployment instructions, architectural references, and incident response guidelines. It enables analysts to deploy, operate, and maintain the SIEM platform efficiently while following standardized cybersecurity practices.

TESTING & VALIDATION

25.1 Testing Overview

Testing and validation were conducted to verify the reliability, functionality, and performance of the AegisAI Enterprise SIEM platform. Each module was individually tested to ensure proper execution of threat detection, event correlation, AI-based incident analysis, IOC lookups, case management, reporting, and security analytics. The testing process confirmed that all integrated components communicate successfully and provide accurate results in real time.

25.2 Testing Environment

Component	Configuration
Operating System	Windows 11 Pro (64-bit)
Frontend	React + TypeScript + Vite
Backend	FastAPI (Python)
Programming Language	Python, TypeScript
Database	SQLite
Browser	Google Chrome
IDE	Visual Studio Code
AI Engine	Gemini API
SIEM Platform	AegisAI Enterprise SIEM
Testing Type	Functional & Integration Testing

25.3 Functional Testing Results

Test Case	Expected Result	Actual Result	Status	Remarks
User Login	Dashboard opens successfully	Passed	PASS	Successful
Dashboard Loading	All widgets displayed	Passed	PASS	Working
Alert Center	Alerts displayed correctly	Passed	PASS	Working
AI Incident Analysis	AI explanation generated	Passed	PASS	Working

MITRE Mapping	ATT&CK techniques loaded	Passed	PASS	Working
Threat Intelligence	IOC records displayed	Passed	PASS	Working
Global Threat Recon	Recon results generated	Passed	PASS	Working
IOC Lookup	IOC search completed	Passed	PASS	Working
Case Management	Case created successfully	Passed	PASS	Working
Audit Reports	Reports exported	Passed	PASS	Working
Security Analytics	Charts displayed	Passed	PASS	Working
User Profile	Profile loaded	Passed	PASS	Working
SIEM Settings	Settings saved	Passed	PASS	Working
Documentation	Documentation opened	Passed	PASS	Working

25.4 Performance Evaluation

Parameter	Observation
Dashboard `Loading	Excellent
User Interface	Responsive
Alert Generation	Real-Time
AI Analysis	Fast
Report Generation	Successful
Memory Usage	Stable
CPU Usage	Normal
Overall Performance	Excellent

25.5 Validation Summary

- All project modules executed successfully.
- Dashboard loaded without runtime errors.
- AI incident analysis generated accurate explanations.
- Security alerts were processed correctly.
- MITRE ATT&CK mapping functioned properly.
- Threat Intelligence and IOC lookup worked successfully.

- Reports were generated successfully.
- Overall platform operated reliably during testing.

25.6 Summary

The testing phase confirmed that the AegisAI Enterprise SIEM platform meets its intended functional requirements. All implemented modules were successfully validated, demonstrating stable performance, reliable AI-assisted threat analysis, and effective enterprise security monitoring capabilities.

PROJECT RESULTS, FUTURE SCOPE & CONCLUSION

26.1 Project Results

The AegisAI Enterprise SIEM project was successfully designed, developed, and validated as an AI-assisted Security Information and Event Management platform. The application integrates threat monitoring, AI-powered incident analysis, MITRE ATT&CK mapping, threat intelligence, IOC investigation, case management, audit reporting, and security analytics into a unified dashboard. During testing, all implemented modules operated successfully and demonstrated stable real-time performance. The project successfully achieved its primary objective of simplifying Security Operations Center (SOC) monitoring through intelligent automation.

26.2 Project Achievements

Achievement	Status
Enterprise SIEM Dashboard Developed	Completed
AI-Based Incident Analysis	Completed
Alert Management Module	Completed
MITRE ATT&CK Integration	Completed
Threat Intelligence Feed	Completed
IOC Lookup Engine	Completed
Case Management System	Completed
Security Analytics Dashboard	Completed
Audit Report Generation	Completed
Professional SOC Documentation	Completed

26.3 Project Outcomes

- Successfully centralized enterprise security monitoring.
- Reduced manual investigation using AI-assisted analysis.

- Improved visibility of enterprise security events.
- Enabled automated threat correlation.
- Simplified incident response workflow.
- Integrated multiple cybersecurity modules into one platform.
- Enhanced analyst productivity.
- Demonstrated enterprise-level SIEM capabilities.

26.4 Future Scope

The AegisAI Enterprise SIEM platform can be extended with advanced cybersecurity technologies to improve detection accuracy, scalability, and enterprise readiness. Future enhancements may include:

-
- Live Splunk Integration
- Wazuh Agent Integration
- Microsoft Sentinel Support
- Elastic Stack Integration
- AI Threat Prediction
- Machine Learning-Based Anomaly Detection
- Automated SOAR Playbooks
- Email & SMS Alerting
- Cloud Deployment (AWS / Azure)
- Docker & Kubernetes Support
- Multi-Tenant SOC Environment
- Mobile Monitoring Application
- Threat Hunting Dashboard
- Malware Sandbox Integration
- Digital Forensics Automation

26.5 Conclusion

The AegisAI Enterprise SIEM project demonstrates the practical implementation of an AI-driven Security Operations Center platform capable of monitoring, analyzing, and responding to cybersecurity

incidents. By combining modern web technologies, artificial intelligence, and enterprise security principles, the platform provides a centralized environment for security monitoring and incident management. This internship project enhanced practical knowledge in SIEM architecture, SOC workflows, AI-assisted threat detection, and cybersecurity automation. The project successfully fulfills its objectives and establishes a strong foundation for future enterprise security solutions.

26.6 Skills Gained

Technical Skills	Professional Skills
SIEM Architecture	Problem Solving
SOC Operations	Analytical Thinking
Incident Response	Documentation
Threat Intelligence	Research Skills
MITRE ATT&CK	Team Collaboration
Python Development	Communication
React Development	Presentation Skills
Cybersecurity Fundamentals	Project Management

REFERENCES

27.1 Books

1. Stallings, W. *Network Security Essentials: Applications and Standards*. Pearson Education.
2. Easttom, C. *Computer Security Fundamentals*. Pearson.
3. Kim, D., & Solomon, M. *Fundamentals of Information Systems Security*. Jones & Bartlett Learning.
4. Shon Harris. *CISSP All-in-One Exam Guide*. McGraw-Hill.

27.2 Official Documentation

1. React Documentation – <https://react.dev/>
2. FastAPI Documentation – <https://fastapi.tiangolo.com/>
3. Python Documentation – <https://docs.python.org/3/>
4. MITRE ATT&CK Framework – <https://attack.mitre.org/>
5. OWASP Foundation – <https://owasp.org/>
6. SQLite Documentation – <https://www.sqlite.org/>
7. Google Gemini API Documentation – <https://ai.google.dev/>

27.3 Security Resources

1. National Institute of Standards and Technology (NIST) Cybersecurity Framework
<https://www.nist.gov/cyberframework>
2. CISA (Cybersecurity & Infrastructure Security Agency)
<https://www.cisa.gov/>
3. VirusTotal Intelligence
<https://www.virustotal.com/>
4. CrowdStrike Falcon Documentation
<https://www.crowdstrike.com/>
5. AWS GuardDuty Documentation
<https://docs.aws.amazon.com/guardduty/>

27.4 Software & Tools Used

Tool	Purpose
React	Frontend Development
TypeScript	Frontend Programming
Python	Backend Development
FastAPI	REST API Development
SQLite	Database
Gemini API	AI Incident Analysis
Visual Studio Code	Development Environment
Google Chrome	Testing
GitHub	Version Control
Windows 11	Operating System

27.5 Citation Note

All technical concepts, cybersecurity frameworks, software documentation, and development references used during the implementation of the AegisAI Enterprise SIEM project were obtained from official documentation, internationally recognized cybersecurity standards, and academic reference materials.